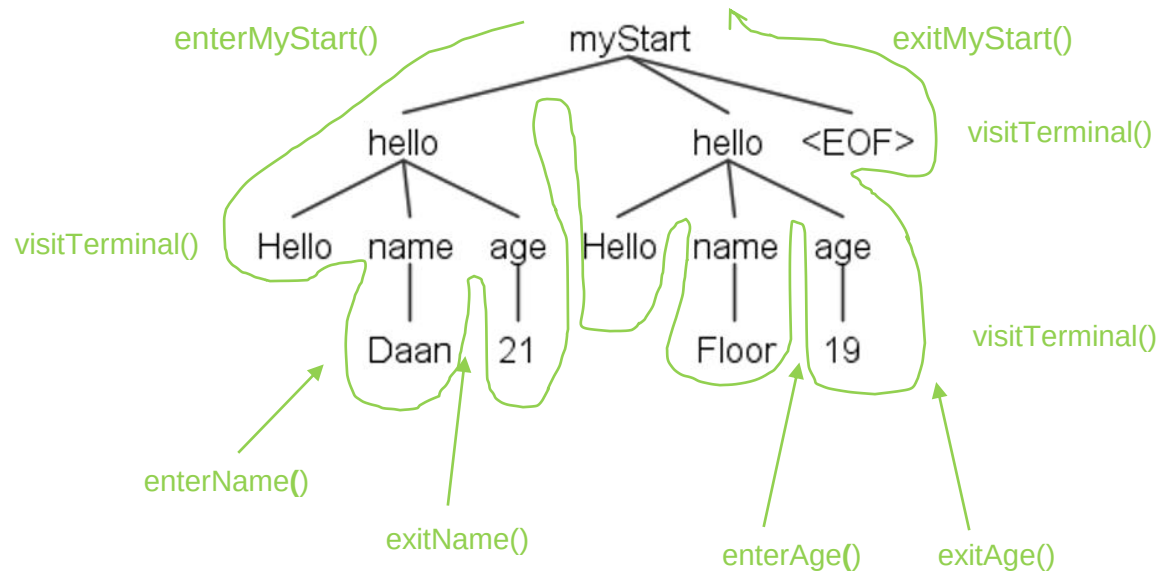


Automata

languages lexers, parsers, interpreters



Recap: Listener Methods



Visitor design pattern

Intent

- **Visitor** is a behavioral design pattern that lets you separate algorithms from the objects on which they operate.

ANTLR: separate grammar from program

Listener vs. Visitor

- Similarities
 - depth-first algorithm
 - provide an easy and standard way to process a parse tree
- Differences
 - **Listener** cannot govern the process of traversing the tree
 - **Visitor** can return values from each visitor function and even stop the visitor together.

Context classes

Grammar

```
myStart      : (hello | goodbye)+ EOF;  
hello        : HELLO name age;  
goodbye      : GOODBYE;  
name         : ID;  
age          : NUMBER;
```

Class HelloContext

```
TerminalNode HELLO()  
NameContext name()  
AgeContext age()
```

Class GoodbyeContext

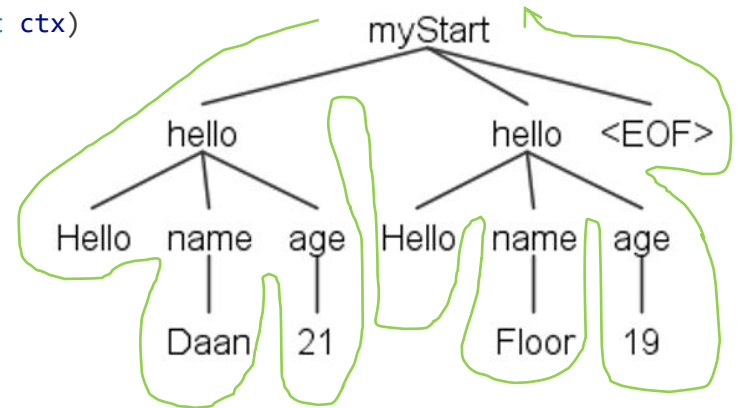
```
TerminalNode GOODBYE()
```

Visit method

APIs

Rest of
application

```
public T  
visitMyStart(MyStartContext ctx)  
  
public T  
visitHello(HelloContext ctx)  
  
public T  
visitName(NameContext ctx)
```



Implementing a Visitor

- Demo